

INTRODUCTION TO INFORMATION & COMMUNICATION TECHNOLOGIES (ICT)

Course Lecture Notes

TOPIC
Introduction to Computers

SUB-TOPIC: Hardware Components — Input, Output, System Unit, Storage & Communication Devices

Item	Details
Subject	Introduction to Information & Communication Technologies (ICT)
Unit / Chapter	Introduction to Computers
Topic of the Day	Hardware Components: Input, Output, System Unit, Storage & Communication Devices
Level	DIT PART-I
Prepared by	Hafiz Muhammad Usman (Lecturer, Computer Science)

1. Learning Objectives

By the end of this lecture, students will be able to:

- Recall and briefly summarize the history and evolution of computers covered previously.
- Identify the major hardware components that make up a computer system.
- Classify hardware devices into Input, Output, System Unit, Storage, and Communication categories.
- Describe the function and examples of common input and output devices.
- Explain the role of the System Unit and its core internal components.
- Differentiate between primary and secondary storage devices.
- Understand how communication devices enable data exchange between computers and networks.

1.1 History and Evolution of Computers (Recap)

Before moving into hardware components, a quick recap of the historical journey of computing covered in the previous lecture:

Era	Period	Highlights
Early Devices	~2400 BC – 1890	Abacus, Napier's Bones, Pascaline, Stepped Reckoner, Jacquard's Loom, Babbage's Engines, Hollerith's Tabulating Machine
1st Generation	1940 – 1956	Vacuum Tubes — ENIAC, UNIVAC I; large, slow, unreliable
2nd Generation	1956 – 1963	Transistors — IBM 1401; smaller, faster, more reliable

Era	Period	Highlights
3rd Generation	1964 – 1971	Integrated Circuits — IBM System/360; Operating Systems introduced
4th Generation	1971 – 2010	Microprocessors — IBM PC, Macintosh; GUI, Internet emerge
5th Generation	2010 – Present	Artificial Intelligence — smartphones, AI assistants, cloud computing

Why This Matters

- Each generation reduced the size and cost of computers while increasing their speed and reliability.
- This evolution set the stage for today's compact, powerful hardware — the focus of this lecture.

1.2 Hardware Components of a Computer

Definition: Hardware

The physical, tangible parts of a computer system that can be seen and touched — including devices used for input, output, processing, storage, and communication.

A complete computer system is built from five broad categories of hardware, each responsible for a stage of the Input–Process–Output–Storage cycle introduced earlier:

- Input Devices — used to enter data and instructions into the computer.
- Output Devices — used to present processed data (information) to the user.
- System Unit — the core casing that houses the CPU, motherboard, and memory.
- Storage Devices — used to save data and programs for current or future use.
- Communication Devices — used to send and receive data between computers and networks.

Category	Function	Examples
Input	Entering raw data/instructions	Keyboard, Mouse, Scanner, Microphone
Output	Presenting processed information	Monitor, Printer, Speakers
System Unit	Housing processing components	CPU, Motherboard, RAM, Power Supply
Storage	Saving data for future use	Hard Disk, SSD, USB Flash Drive, Optical Disc
Communication	Exchanging data over networks	Modem, NIC, Router, Wi-Fi Adapter

2. Input Devices

Definition: Input Device

Any hardware component that allows a user to enter data, commands, or instructions into a computer for processing.

Device	Type	Function
Keyboard	Text and Command Entry	Enters letters, numbers, and symbols; issues shortcut commands.
Mouse	Pointing Device	Controls the on-screen pointer; used to click, drag, and select.
Scanner	Image/Document Capture	Converts printed documents or images into digital form.
Microphone	Audio Input	Captures sound and voice for recording or voice commands.
Webcam	Video Input	Captures live video for calls, recording, and streaming.
Touchscreen	Direct Touch Input	Allows direct interaction with the display using fingers or a stylus.
Barcode Reader	Optical Data Capture	Scans barcodes to quickly enter product or item data.
Joystick	Gaming/Control Input	Used for gaming and controlling simulations or machinery.

Why This Matters

- Input devices are the computer's only way of receiving raw data — without them, no processing can begin, following the GIGO principle from Week 1.

3. Output Devices

Definition: Output Device

Any hardware component that presents processed data (information) from the computer to the user, in visual, audio, or printed form.

Device	Type	Function
Monitor	Visual (Soft Copy)	Displays text, images, and video on screen.
Printer	Visual (Hard Copy)	Produces a physical, paper copy of digital documents.
Speakers	Audio	Converts digital audio signals into audible sound.
Projector	Visual (Large Display)	Projects the computer's display onto a large screen or wall.
Headphones	Audio (Personal)	Delivers audio output privately to an individual user.
Plotter	Visual (Large Format)	Prints high-quality large-scale drawings, e.g., architectural plans.

4. The System Unit

Definition: System Unit

The main casing of a computer that houses the core electronic components responsible for processing data, including the CPU, motherboard, memory, and power supply.

Component	Function
CPU (Processor)	The 'brain' of the computer; fetches, decodes, and executes instructions.
Motherboard	The main circuit board connecting all internal components together.
RAM (Primary Memory)	Temporary, volatile memory that holds data and instructions currently in use.
ROM	Permanent, non-volatile memory storing essential startup instructions (BIOS/firmware).
Power Supply Unit (PSU)	Converts electrical power from the outlet into usable power for internal parts.
Expansion Slots/Cards	Allow additional components (e.g., graphics cards, sound cards) to be added.
Cooling Fan/Heat Sink	Dissipates heat generated by the CPU and other components.

Why This Matters

- The System Unit is where the actual 'Process' stage of the IPO Cycle takes place — it is the electronic equivalent of Babbage's 'Mill' from the Analytical Engine.

5. Storage Devices

Definition: Storage Device

A hardware component used to record (store) and retrieve data, either temporarily (primary storage) or permanently (secondary storage).

5.1 Primary vs. Secondary Storage

Aspect	Primary Storage	Secondary Storage
Volatility	Volatile (data lost when power off)	Non-volatile (data retained without power)
Speed	Very fast access	Slower than primary storage
Capacity	Relatively small (GBs)	Large (GBs to TBs)
Examples	RAM, Cache Memory	Hard Disk, SSD, USB Drive, Optical Disc
Purpose	Holds data currently being processed	Stores data and programs long-term

5.2 Common Secondary Storage Devices

Device	Technology	Description
Hard Disk Drive (HDD)	Magnetic	Large-capacity, cost-effective storage using spinning magnetic platters.
Solid State Drive (SSD)	Flash-based	Fast, durable storage with no moving parts; increasingly standard.
USB Flash Drive	Flash-based	Small, portable storage for transferring files between devices.
Optical Disc (CD/DVD/Blu-ray)	Optical	Stores data using laser-read patterns; used for media and backups.
Memory Card (SD Card)	Flash-based	Compact storage used in cameras, phones, and portable devices.
Cloud Storage	Network-based	Stores data on remote servers, accessible via the Internet.

6. Communication Devices

Definition: Communication Device

Hardware that enables a computer to send and receive data to and from other computers or devices, typically over a network or the Internet.

Device	Type	Function
Modem	Modulator-Demodulator	Converts digital data to analog signals (and back) for transmission over telephone or cable lines.
Network Interface Card (NIC)	Wired Networking	Enables a computer to connect to a wired local network (Ethernet).
Wi-Fi Adapter	Wireless Networking	Enables a computer to connect to wireless networks.
Router	Network Routing	Directs data traffic between networks and connects devices to the Internet.
Bluetooth Adapter	Short-range Wireless	Enables short-range wireless communication between nearby devices.
Hub/Switch	Network Connectivity	Connects multiple devices within a local area network (LAN).

Why This Matters

- Communication devices connect the isolated 'stand-alone' computers of earlier generations into the networked, Internet-driven world of the Fourth and Fifth Generations.

7. Comparative Summary of Hardware Categories

Category	Primary Role	Data Flow	Examples
Input	Enter data/instructions	Human → Computer	Keyboard, Mouse, Scanner
Output	Present information	Computer → Human	Monitor, Printer, Speakers
System Unit	Process data	Internal	CPU, Motherboard, RAM
Storage	Save data	Internal/External	HDD, SSD, USB Drive
Communication	Exchange data	Computer ↔ Network	Modem, Router, NIC

8. Key Terms & Glossary

Term	Meaning
Peripheral	Any hardware device connected to a computer that is not part of the core system unit (e.g., keyboard, printer).
Port	A physical connection point on the system unit used to attach peripherals (e.g., USB port).
Driver	Software that allows the operating system to communicate with a hardware device.
Bandwidth	The maximum rate of data transfer across a communication channel or network.
Plug and Play	A feature that allows a device to be automatically recognized and configured when connected.
IoT (Internet of Things)	A network of physical devices embedded with sensors and connectivity to exchange data.

9. Summary

Key Takeaways

- Computer hardware is organized into five categories: Input, Output, System Unit, Storage, and Communication.
- Input devices capture data; output devices present processed information to the user.
- The System Unit — housing the CPU, motherboard, and memory — is where actual processing occurs.
- Storage devices are divided into fast, volatile primary storage (RAM) and slower, permanent secondary storage (HDD/SSD).
- Communication devices (modem, NIC, router, Wi-Fi) connect computers into networks and the Internet.
- Together, these hardware categories implement the Input–Process–Output–Storage cycle introduced in Week 1.

10. Review Questions (For Practice / Assignment)

- List and briefly describe five common input devices, other than the keyboard and mouse.
- Differentiate between primary storage and secondary storage with suitable examples.
- Explain the role of the CPU and motherboard within the System Unit.
- Construct a table comparing HDD, SSD, and Cloud Storage on the basis of speed, cost, and portability.
- Discuss how communication devices such as a router and NIC enable networking.
- In your opinion, how have touchscreens changed the way input devices are designed for modern devices?